

2-Year Roadmap to Become an Embedded Engineer

This roadmap is designed for students who want to become Embedded Engineers within 2 years, focusing on C programming, microcontrollers, RTOS, embedded Linux, and project development.

■ YEAR 1 — FOUNDATIONS + MINI PROJECTS

Semester 1 (Months 1–4): C Programming + Electronics Basics

- Learn: C programming (pointers, arrays, structures), Digital Electronics (gates, flip-flops, counters).
- Tools: TinkerCAD / Proteus for simulation.
- Resources: Jenny's Lectures (C), Neso Academy (Digital).
- Practice: 100+ problems on HackerRank.

Semester 2 (Months 5–8): Microcontrollers + Embedded C

- Learn: 8051/AVR, timers, interrupts, UART, SPI, I2C.
- Tools: Keil, Proteus.
- Projects: LED Blinking, Digital Thermometer, Bluetooth Control.
- Resources: Neso Academy (8051), FastBit STM32 course.

■ YEAR 2 — ADVANCED CONCEPTS + PROJECTS + PLACEMENT PREP

Semester 3 (Months 9–14): Advanced Embedded + RTOS

- Learn: ARM Cortex-M (STM32), FreeRTOS (tasks, queues, semaphores).
- Projects: RTOS Sensor Monitor, Home Automation, CAN Bus Comm.
- Resources: FastBit RTOS course, NPTEL Embedded Systems.

Semester 4 (Months 15–20): Embedded Linux + IoT

- Learn: Linux commands, GPIO in Raspberry Pi, MQTT, Device Drivers.
- Projects: IoT Smart Energy Meter, Cloud Temp Logger.
- Resources: Derek Molloy (Linux), Jonathan Corbet (Linux Drivers).

Semester 5 (Months 21–24): Final Project + Internship + Placement Prep

- Major Projects: RTOS-based Automation, IoT Cloud System.
- Internships: NIELIT, CDAC, ISRO, Bosch, Qualcomm.
- Prepare: Embedded C, RTOS, Communication, and interview questions.

■ Additional Skills to Learn

Skill	Tools / Platforms	Benefit
PCB Design	KiCAD / Altium	Build and test your own circuits

Python Basics	YouTube / freeCodeCamp	IoT and scripting
MATLAB / Simulink	NPTEL	Control system modeling
FPGA Basics	Verilog / Vivado	Optional for VLSI field

■ Recommended Certifications

- NIELIT Calicut – Embedded Systems / IoT Design
- AICTE Internship Portal – Embedded, IoT, or VLSI
- NPTEL – Embedded Systems / Microcontrollers
- CDAC – Advanced Embedded / RTOS

■ Target Companies

Bosch, NXP, Qualcomm, Intel, Texas Instruments, CDAC, DRDO, ISRO, VVDN, Emertxe, Sasken, Tata Elxsi.